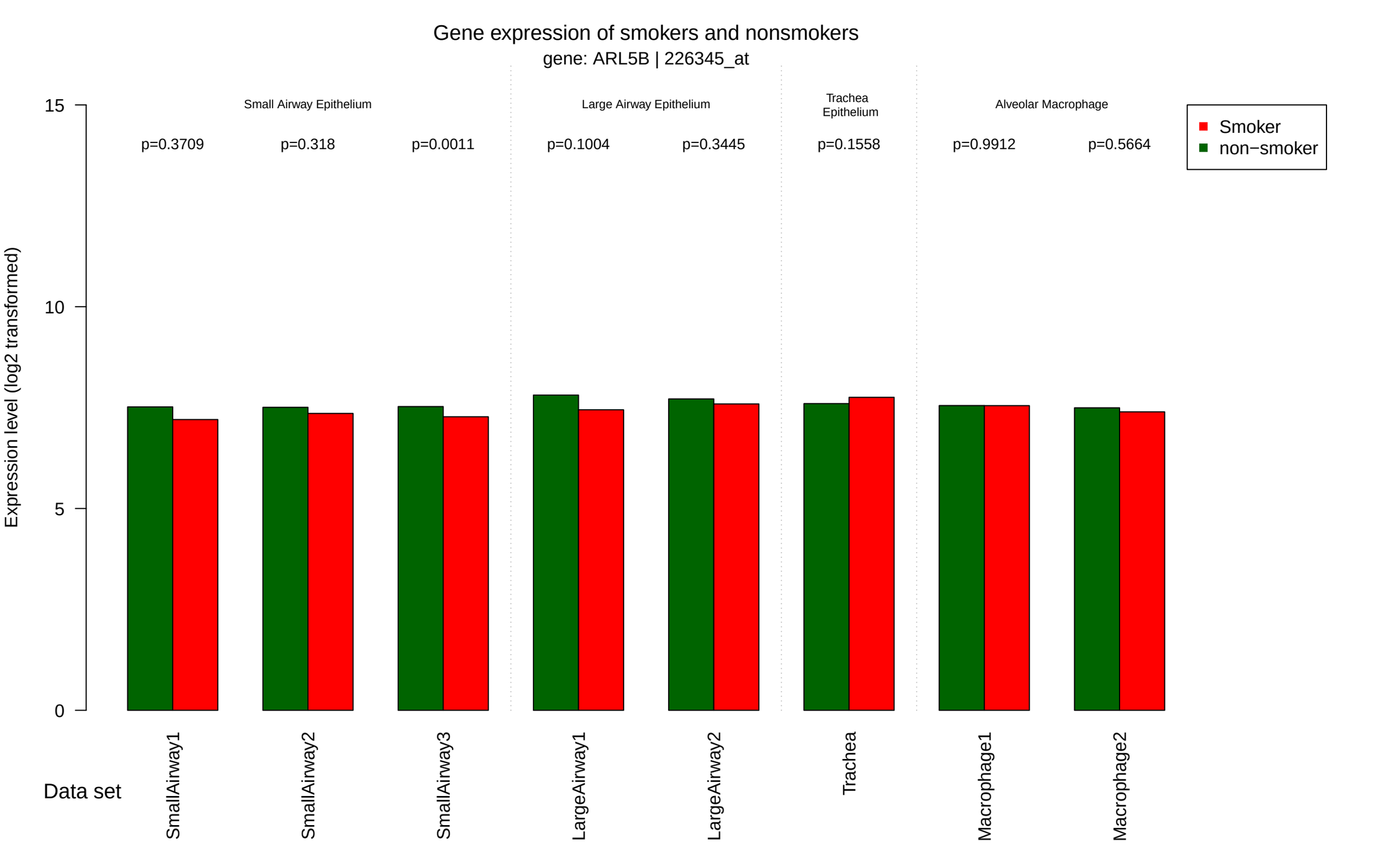
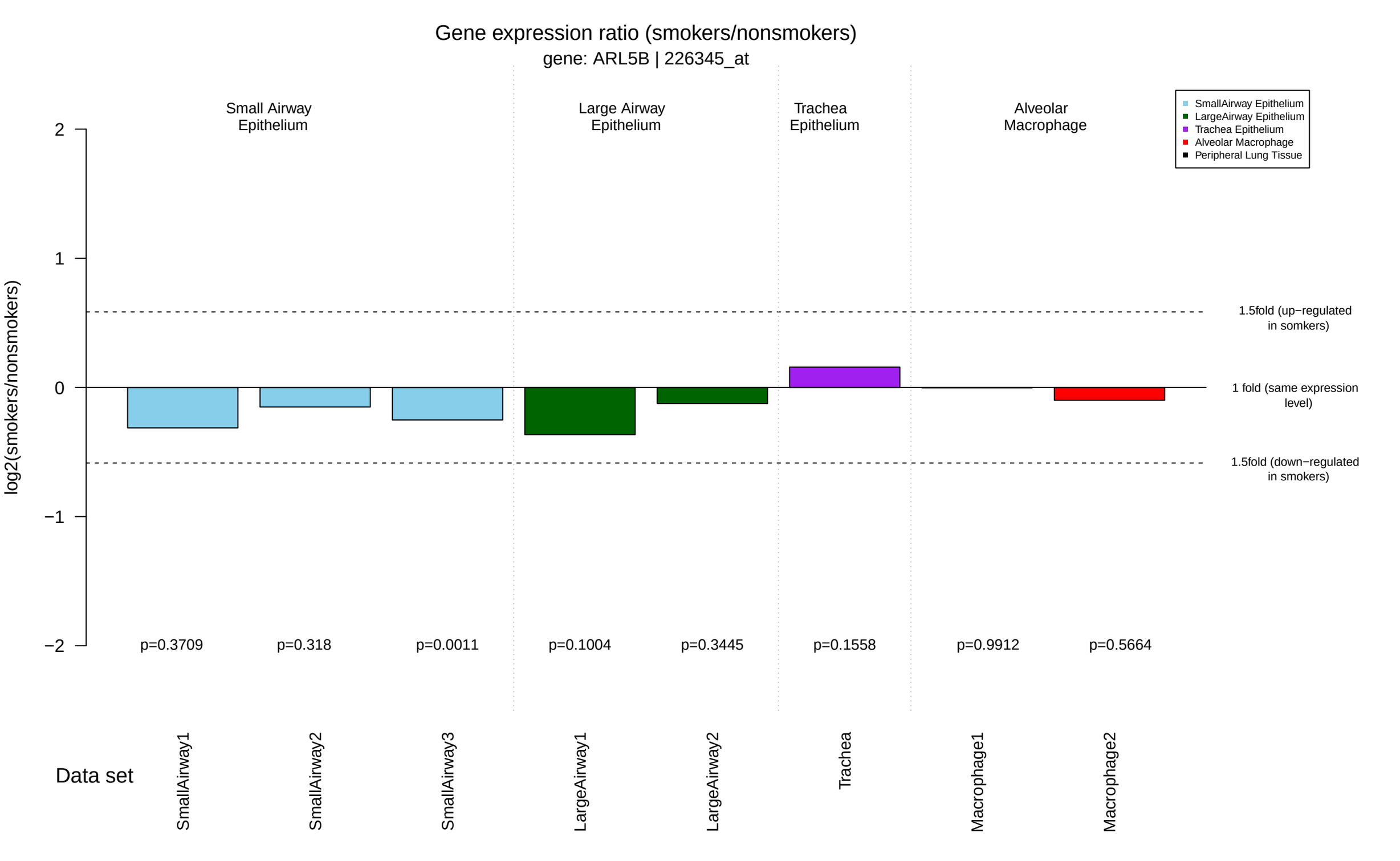
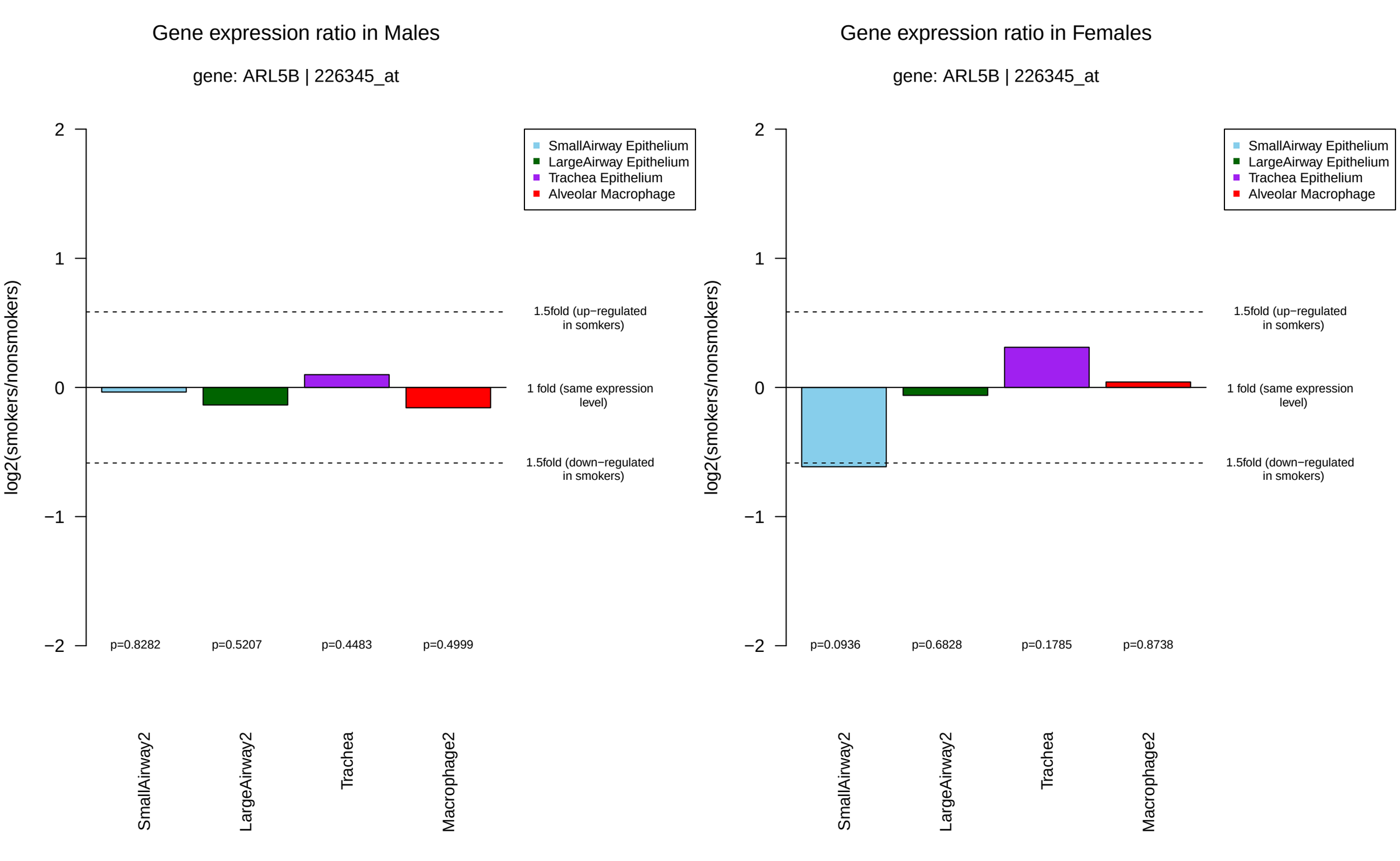
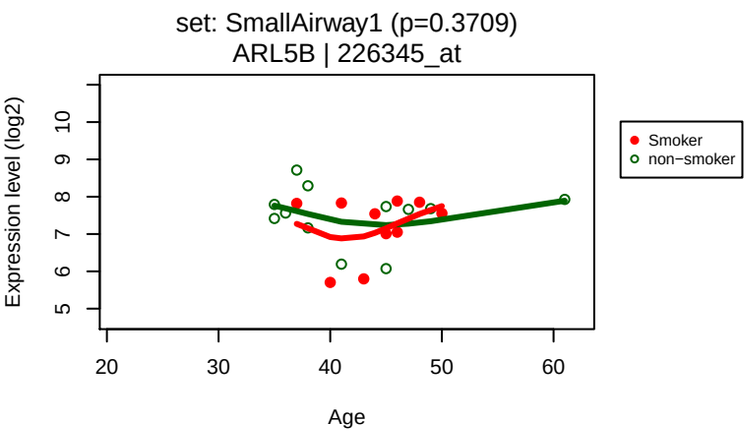
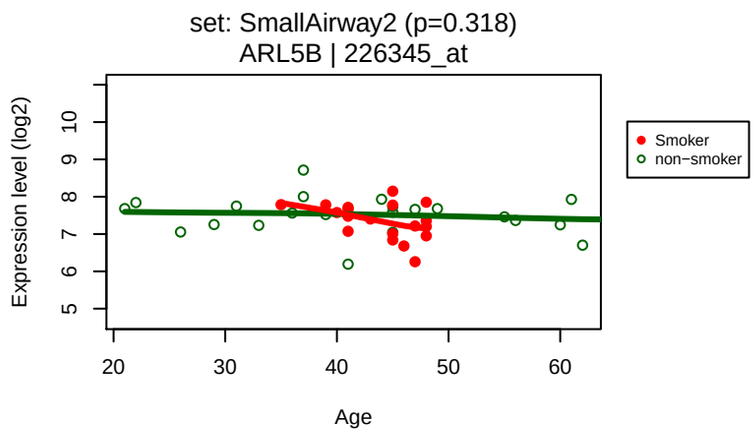
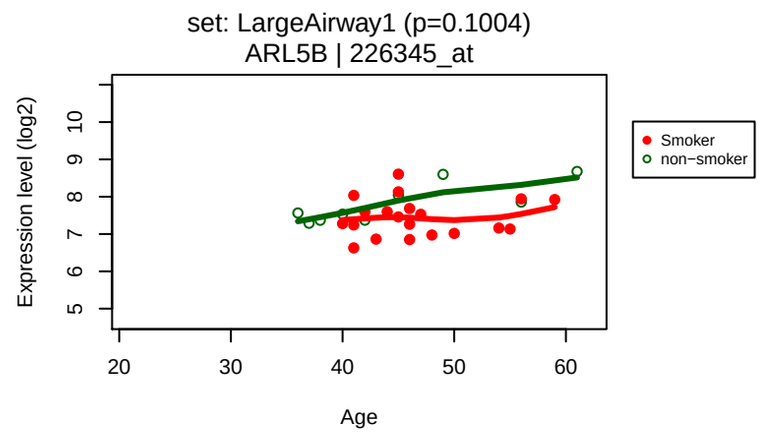

Gene expression of smokers and nonsmokers
gene: ARL5B | 226345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3709
p=0.318
p=0.0011
p=0.1004
p=0.3445
p=0.1558
p=0.9912
p=0.5664
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

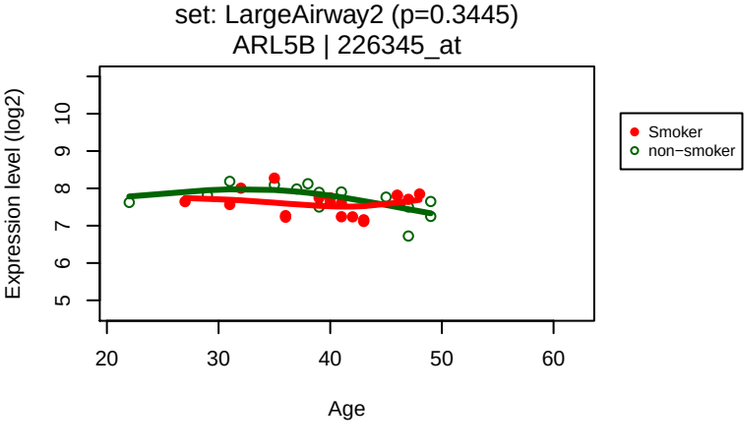

Gene expression ratio (smokers/nonsmokers)
gene: ARL5B | 226345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3709
p=0.318
p=0.0011
p=0.1004
p=0.3445
p=0.1558
p=0.9912
p=0.5664
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

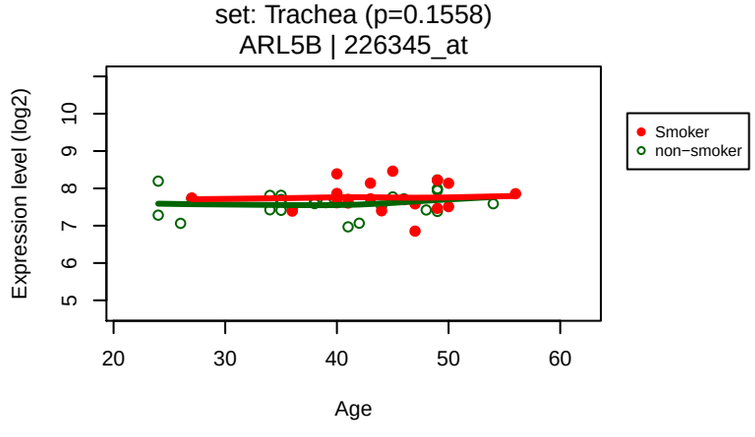

Gene expression ratio in Males
gene: ARL5B | 226345_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8282
p=0.5207
p=0.4483
p=0.4999
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL5B | 226345_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0936
p=0.6828
p=0.1785
p=0.8738
SmallAirway2
LargeAirway2
Trachea
Macrophage2

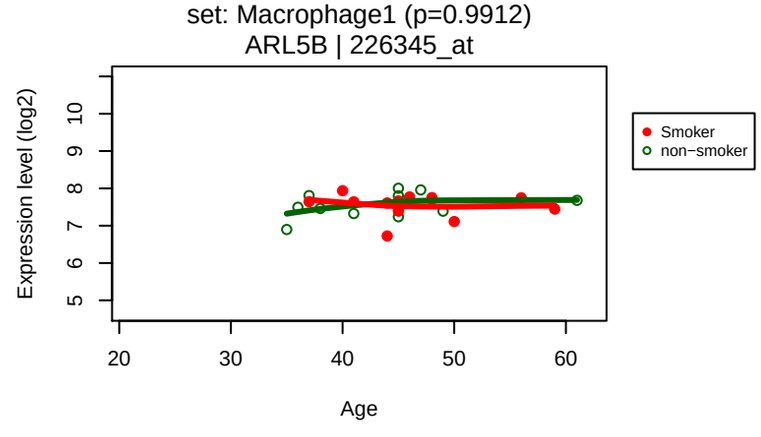

set: SmallAirway1 (p=0.3709)
ARL5B | 226345_at
Expression level (log2)
Age
Smoker
non−smoker

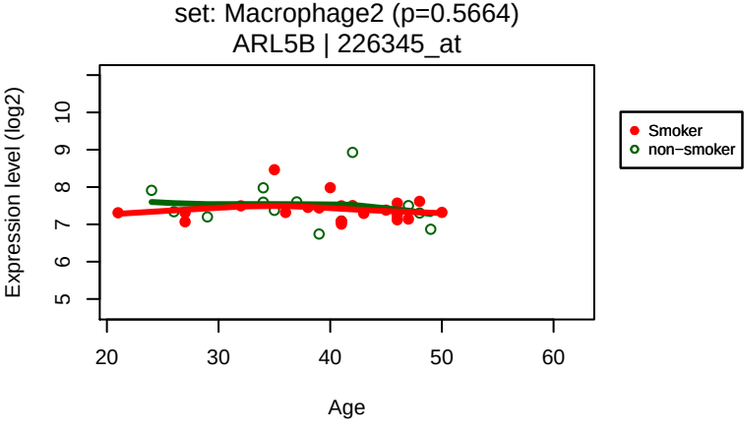

set: SmallAirway2 (p=0.318)
ARL5B | 226345_at
Expression level (log2)
Age
Smoker
non−smoker

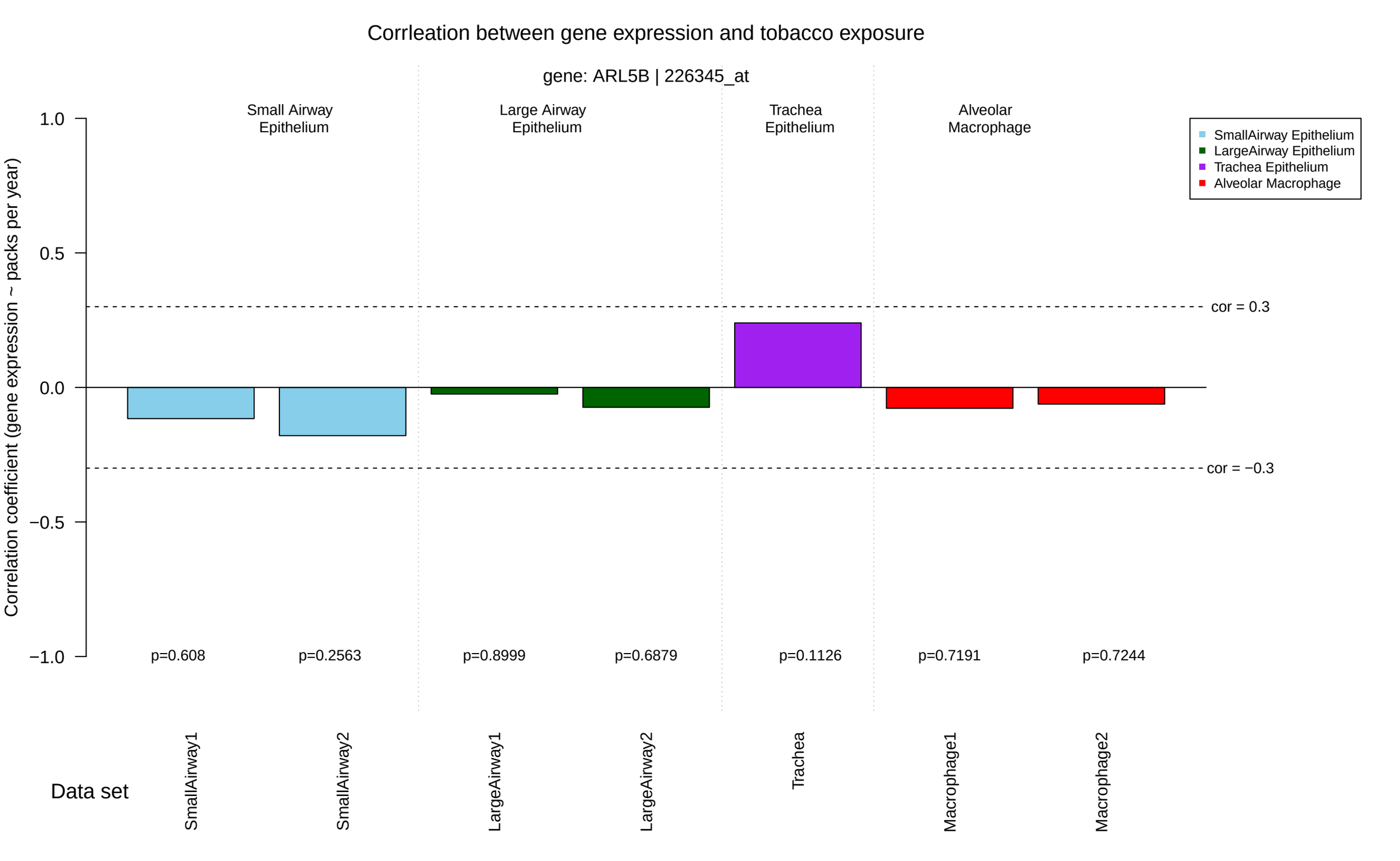

set: LargeAirway1 (p=0.1004)
ARL5B | 226345_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3445)
ARL5B | 226345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1558)
ARL5B | 226345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9912)
ARL5B | 226345_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5664)
ARL5B | 226345_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL5B | 226345_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.608
p=0.2563
p=0.8999
p=0.6879
p=0.1126
p=0.7191
p=0.7244
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2